



MediQuick AI

An Advanced Multimodal Medical Triage Assistant

Mid-Semester Progress Review · June 2026



The Problem

*“How urgent is this — **and who should I see?**”*

1

Noise, not clarity

Search results are overwhelming and often alarming, not reassuring.

2

Cost & access

Booking a consultation for every minor concern is impractical and expensive.

3

Wrong question

People need to know the urgency and the right specialist — not a self-diagnosis.



The Approach

MediQuick AI triages — it does not diagnose. Describe symptoms in chat, optionally add a photo, and receive a structured, urgency-ranked assessment.

Multimodal

Understands text and images together via a Vision-Language Model.

Structured JSON

Every reply follows a strict, validated six-field schema.

Three urgency tiers

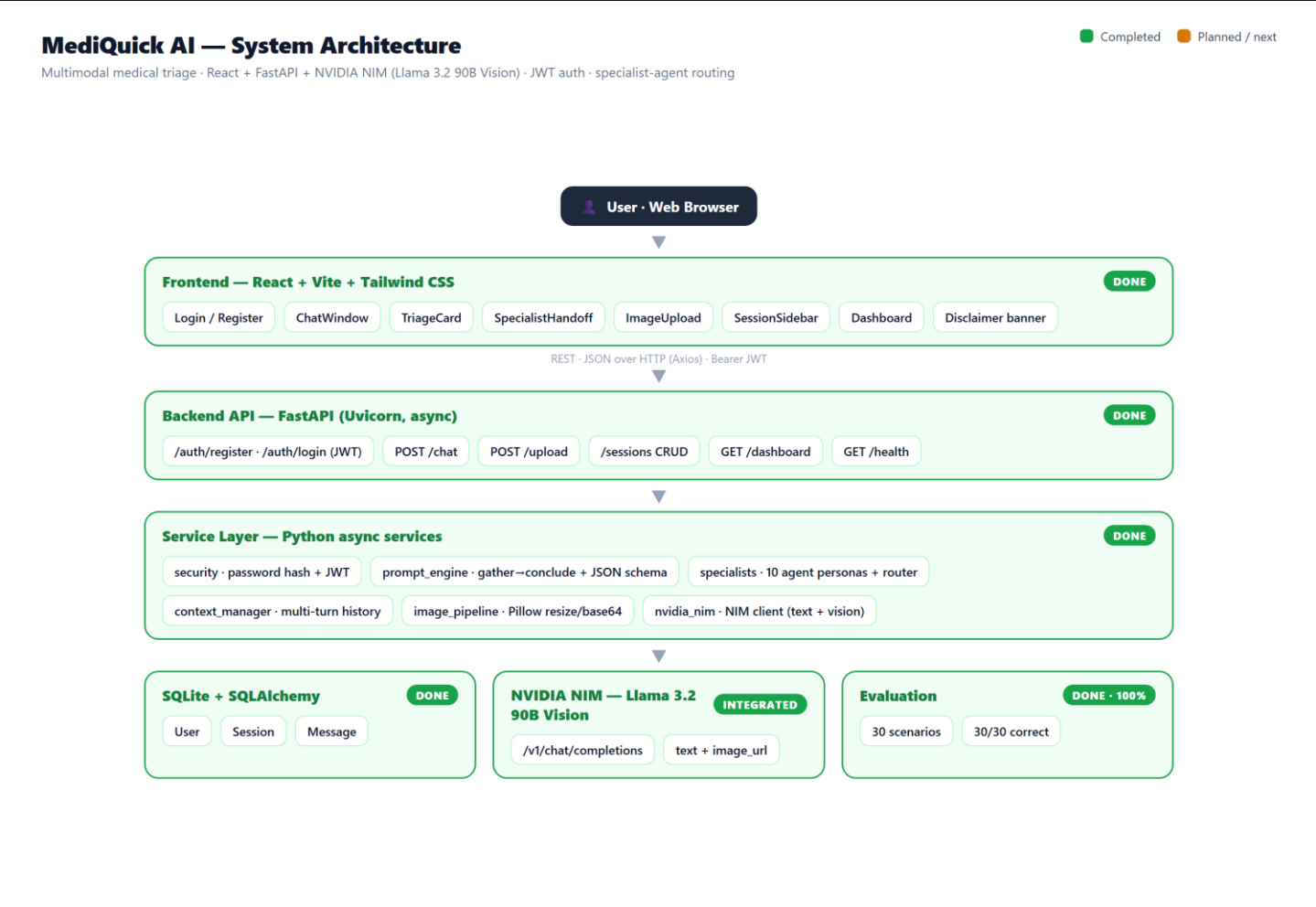
Self-care · consult a doctor · seek emergency care.

Safe by design

Emergency override and a disclaimer on every response.



System Architecture



React + FastAPI + NVIDIA NIM · JWT auth · specialist-agent routing — all layers complete and validated.

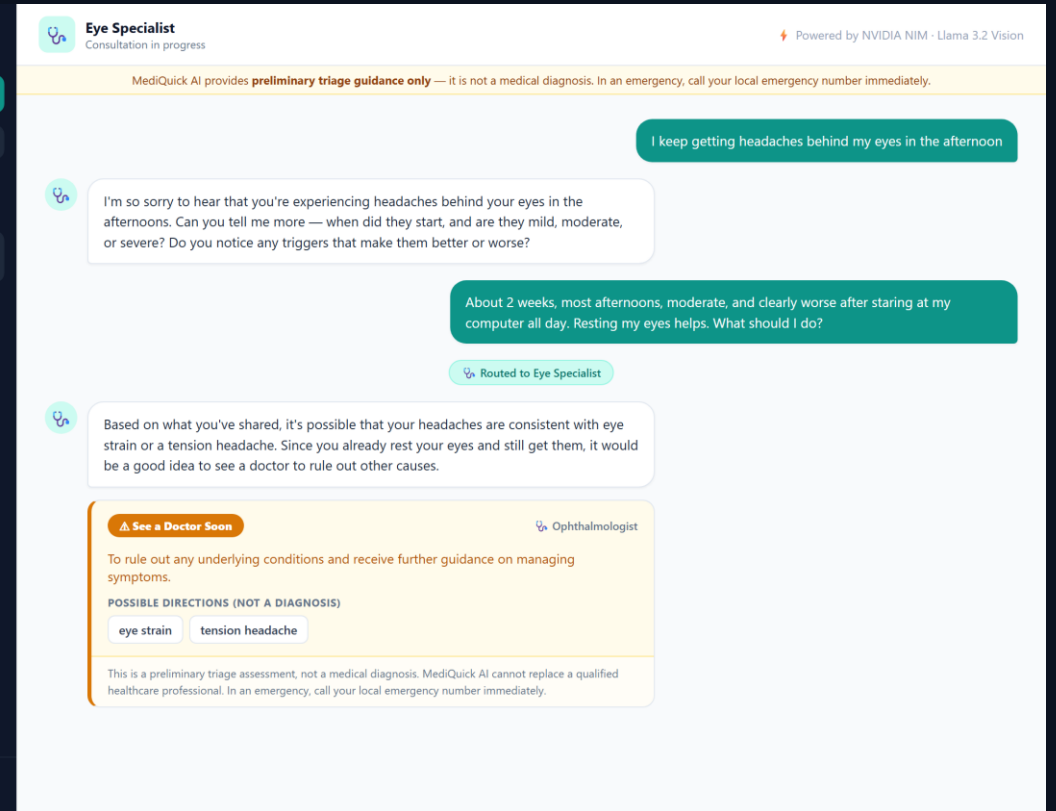
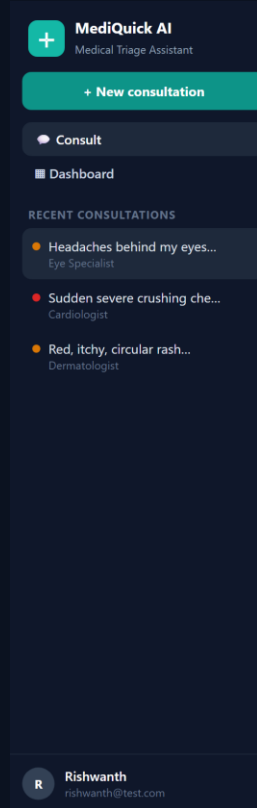
+ Demo • Doctor-Style Consultation

1 User describes a symptom in plain language.

2 The assistant asks focused follow-up questions.

3 It hands off to the right specialist agent.

4 It concludes with a colour-coded triage card.



Real consultation: General Physician → Eye Specialist handoff, concluding with a triage card.



Specialist Routing & Dashboard

+

MediQuick AI

Medical Triage Assistant

+ New consultation

Consult

Dashboard

RECENT CONSULTATIONS

Headaches behind my eyes...

Eye Specialist

Sudden severe crushing che...

Cardiologist

Red, itchy, circular rash...

Dermatologist

R

Rishwanth

rishwanth@test.com

Your health dashboard

A summary of every consultation you've concluded. Tap any card to reopen the conversation.

Consultations

3

Concluded

3

Self-care

0

Need a doctor

3

RESOLUTIONS

See a Doctor Soon

Eye Specialist

I keep getting headaches behind my eyes ...

To rule out any underlying conditions and receive further guidance on managing symptoms.

eye strain tension headache

See a Doctor Soon

Dermatologist

I have had a red, itchy, circular rash o...

The rash is persistent and may require further evaluation to rule out other conditions.

ringworm eczema contact dermatitis

Seek Emergency Care

Cardiologist

I have sudden severe crushing pain in th...

Severe chest pain, sweating, and shortness of breath are indicative of a potential cardiac emergency.

Myocardial Infarction (Heart Attack) Acute Coronary Syndrome

10 agents

General Physician routes to the right specialist

Context carried

Full history follows the handoff

Photos in-memory

Validated, resized, never saved to disk

Dashboard: every concluded consultation summarised by urgency and specialist — no medication content.



Evaluation · 30 Structured Scenarios

100%

urgency-classification accuracy

0 unsafe under-triage cases

Dermatology	10 / 10
General medicine	10 / 10
Minor injury	10 / 10
Overall	30 / 30

Compared against standard triage reference guidelines, across all three urgency tiers.

Progress at the Milestone

88%

of the project complete

full app + evaluation — only polish remains

Completed

- ✓ FastAPI backend + JWT auth
- ✓ NVIDIA NIM client (text + vision)
- ✓ Prompt engine, gather→conclude
- ✓ 10 specialist agents + routing
- ✓ React UI: chat, card, sidebar
- ✓ Dashboard of resolutions
- ✓ Evaluation — 30/30 (100%)

Remaining

- Error-handling & UI polish
- Automated backend tests
- Responsive-layout review
- Developer README & handover

What's Next

1

Polish error handling Harden empty / loading / error states across the interface.

2

Automated tests A short test suite for the backend services.

3

Responsive review Tune the layout for smaller screens.

4

Handover docs Developer README and setup guide for the team.



Thank You

MediQuick AI provides preliminary triage guidance only — not a medical diagnosis.

